

BrainRiskNet: Dual Head CNN–ConvLSTM for Brain Tumor MRI Classification with Uncertainty Aware Severity Assessment

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Abstract—Automated brain tumor MRI classification can support screening and triage, but most classifiers provide only a class label and little information about severity or prediction uncertainty. We present BrainRiskNet, a dual-head CNN–ConvLSTM framework for four-class tumor classification, auxiliary three-class severity prediction, and entropy-based uncertainty flagging in a single inference-time pipeline. During BrainRiskNet training, each input image is cropped, resized to 224×224 pixels, normalized, and augmented online; by contrast, the baseline model is trained only on clean images. Evaluated on a fixed held-out set of 1600 images, BrainRiskNet reaches 91.12% accuracy, 91.50% precision, 91.12% recall, and 90.98% weighted F1 for tumor classification, outperforming the compact CNN baseline by 2.74 percentage points through the combined effect of architecture, augmentation, and class weighting. The auxiliary risk head reaches 92.50% accuracy and 92.33% weighted F1, and 64 test cases (4.0%) are flagged as uncertain. Because the risk labels are derived from class-level severity proxies rather than patient outcomes, the risk output should be interpreted as a research-oriented severity indicator rather than a validated clinical endpoint.

Index Terms—brain tumor MRI classification, CNN–ConvLSTM, multitask learning, uncertainty quantification, severity stratification, predictive entropy, medical image analysis, deep learning

I. INTRODUCTION

Accurate classification of brain tumor magnetic resonance imaging (MRI) scans plays an important role in clinical decision-making and treatment management, as the major diagnostic categories vary considerably in malignancy risk, referral urgency, and follow-up care requirements. Compared with pituitary adenomas and most meningiomas, gliomas generally have a less favorable prognosis. Consequently, assigning a high-risk lesion to a lower-severity category may delay specialist evaluation and subsequent clinical management [1], [2]. However, many current deep learning methods focus mainly on improving per-class accuracy and hence treat classification errors as equally important, despite their different implications for clinical decision-making across healthcare systems [3]–[5].

This creates two practical challenges. First, it assumes that all classification mistakes have the same impact, even though their clinical consequences can vary significantly. For example,

failing to detect a potential glioma is generally far more serious than misidentifying one lower-risk tumor type as another. Second, models that provide only a final class prediction do not indicate how confident they are in that prediction, making it difficult to recognize cases that may require additional expert assessment. Therefore, in a clinical workflow, a system capable of identifying uncertain predictions and directing such cases for further review may be more useful than one that reports classification accuracy alone.

BrainRiskNet is designed to address these limitations through a dual-head architecture that performs tumor classification, estimates severity-based risk, and identifies uncertain predictions. This work makes three key contributions. First, it presents a reproducible four-class CNN–ConvLSTM model trained from scratch on a balanced public split, achieving a 2.74 percentage-point accuracy gain over a compact CNN baseline. Second, it adds an auxiliary three-class risk branch based on the WHO CNS5 severity hierarchy, reaching 92.50% accuracy. Third, it introduces a post-processing severity score that uses predictive entropy to detect uncertain cases without extra annotations or retraining.

II. RELATED WORK

Research on brain tumor MRI classification has explored several directions, including transfer learning, custom CNN architectures, and hybrid classifiers [4], [5]. Within the same dataset family—or on related public benchmarks—Chauhan et al. adopted a different structural strategy with BT-CNN. Instead of a single-stage design, BT-CNN uses a balanced binary tree to decompose tumor-type classification hierarchically [6]. Rasheed et al. reported 85.97% accuracy using an InceptionV3 model in a three-class setup [7], whereas Rasool et al. achieved 98.1% with a GoogLeNet–SVM hybrid approach [8].

Because these studies differ in dataset version, number of classes, augmentation strategy, and use of ImageNet-pretrained weights, their reported results are difficult to compare directly.

Multitask learning, which enables a shared representation to support multiple prediction targets, has also been shown to improve generalization across a wide range of vision tasks [9].

TABLE I
EXPERIMENTAL DATA SPLIT USED IN THIS STUDY.

Split	Glioma	Meningioma	No tumor	Pituitary	Total
Train	1120	1120	1120	1120	4480
Validation	280	280	280	280	1120
Test	400	400	400	400	1600

This idea motivates the dual-head design of BrainRiskNet, in which tumor classification and auxiliary risk prediction are produced by the same convolutional and recurrent backbone.

Uncertainty quantification is especially important in medical imaging because a useful system should recognize difficult cases rather than assign unreliable predictions with high confidence. Bayesian dropout and related uncertainty methods provide broader foundations for estimating model uncertainty in deep networks [10]. In this study, predictive entropy is used as a practical one-pass uncertainty score that does not require architectural changes. In medical imaging, Jungo et al. incorporated uncertainty into brain tumor segmentation and survival prediction, whereas Ghesu et al. demonstrated that explicitly modeling classification uncertainty can improve chest radiograph assessment [11], [12]. In this dataset setting, we did not identify prior work that combines entropy-based uncertainty flagging with a severity-stratified auxiliary output within a single pipeline.

III. DATASET AND PREPROCESSING

This study uses the Brain Tumor MRI Dataset [13], which includes 7,200 MRI images evenly divided into four classes—glioma, meningioma, pituitary tumor, and no tumor. In the original split, each class has 1,400 images for training and 400 images for testing. A further stratified split of the training folder yields 4480 training images and 1120 validation images, while the fixed test set contains 1600 images. Table I summarizes the final split used for model development and testing.

Each image is cropped around the brain region, resized to 224×224 pixels, and normalized to the unit interval. During BrainRiskNet training, images are also subjected to online augmentation through horizontal and vertical flips, rotations within $\pm 15^\circ$, zoom up to 10%, and translations of up to 10% along each axis. The auxiliary three-class risk labels are derived from the tumor class: no tumor is mapped to low, meningioma and pituitary are mapped to medium, and glioma is mapped to high. These labels are used only as a severity-oriented research target and not as clinical outcome labels. The mapping follows the malignant-potential ordering in WHO CNS5 [1].

IV. PROPOSED ARCHITECTURE

A. Network Design

Each of the baseline model’s four blocks follows the same sequence: convolution, batch normalization, rectified linear unit activation, max pooling, and dropout. Filter counts increase across the blocks from 32 to 64, 128, and 256, after which global average pooling feeds a 256-unit dense layer. Keeping this reference network relatively shallow was intentional, so that comparisons with BrainRiskNet would stay well controlled.

The proposed model, BrainRiskNet, combines three convolutional blocks with a ConvLSTM refinement stage. It follows the framework described in [14] and builds on the long short-term memory architecture introduced in [15]. Although the input is a single two-dimensional image rather than a temporal sequence, the feature map produced by the third convolutional block is reshaped into a singleton pseudo-sequence before being passed to the ConvLSTM. This design allows gated recurrent updates to refine structured spatial representations, enabling the ConvLSTM to capture spatial context rather than genuine temporal dynamics.

The convolutional backbone uses filter sizes of 32, 64, and 128 across its three blocks. Meanwhile, the recurrent refinement stage includes four ConvLSTM blocks with filter sizes of 192, 192, 128, and 128. The shared representation is then passed through 512-unit and 256-unit dense layers before branching into a four-class tumor-classification head and a three-class risk head. The full model contains 7 832 007 parameters, compared with 457 156 for the baseline network. Fig. 1 and Table II summarize the proposed design.

B. Optimization and Regularization

Training uses Adam with early stopping and class-dependent sample weights of 1.8, 1.4, 0.7, and 0.7 for glioma, meningioma, no tumor, and pituitary, respectively. This weighting emphasizes the higher-risk glioma and meningioma classes. The multitask objective is the tumor-classification cross-entropy plus a 0.5-weighted cross-entropy term for the risk head [9]. Both models are trained for up to 100 epochs with a global batch size of 64. The baseline uses clean images, whereas BrainRiskNet uses online augmentation and a lower initial learning rate to stabilize the larger recurrent model.

All convolutional blocks use batch normalization after convolution and before the rectified linear activation. The CNN baseline uses dropout rates of 0.30, 0.30, 0.40, and 0.40 across its four convolutional blocks, followed by 0.50 dropout after the 256-unit dense layer. BrainRiskNet uses dropout rates of 0.25, 0.25, and 0.30 in the convolutional feature extractor, 0.15 after each ConvLSTM block, and 0.30 after flattening before the shared dense representation. All convolutional and dense layers use L2 regularization with coefficient 0.0001. The learning-rate schedules also differ: BrainRiskNet uses Adam with cosine decay from an initial learning rate of 0.0005, whereas the baseline starts at 0.001 and uses ReduceLROnPlateau after validation stagnation.

The full experimental pipeline ran in TensorFlow 2.19.0, with MirroredStrategy distributing training across two NVIDIA Tesla T4 GPUs. Metrics were logged at each epoch, and end-to-end training time for BrainRiskNet was approximately 44.7 min.

C. Post-Hoc Severity Scoring and Evaluation

The post-hoc scoring module is used only during inference and is separate from the auxiliary three-class risk head used during training. Let $\mathbf{p}$ be the four-class softmax output. We compute a continuous severity score as $s = \mathbf{p}^\top \mathbf{w}$. The weights

BRAINRISK-NET DUAL-HEAD ARCHITECTURE

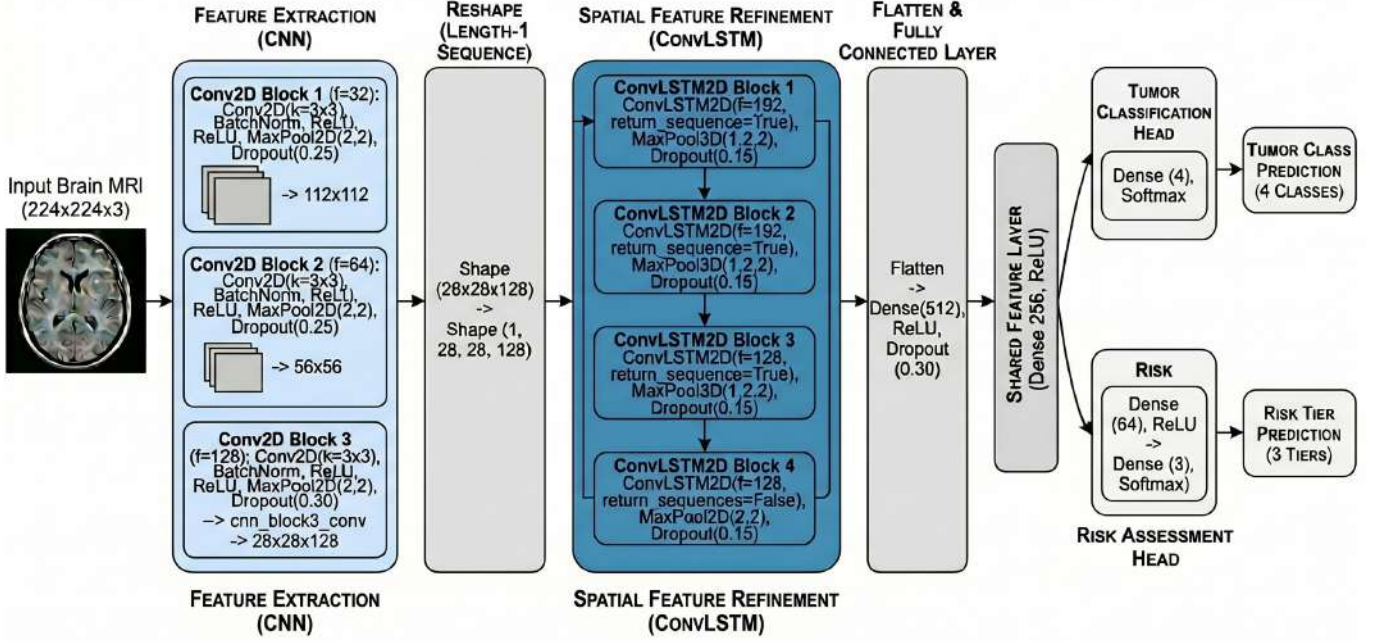


Fig. 1. Architecture of the proposed BrainRiskNet framework. A CNN feature extractor feeds a singleton sequence ConvLSTM refinement stage, followed by shared dense features and separate tumor classification and risk assessment heads.

TABLE II
COMPACT SUMMARY OF THE BASELINE NETWORK AND BRAINRISKNET.

Stage	Baseline network	BrainRiskNet
Input	224 × 224 × 3 image	Same input and preprocessing
Feature extractor	Four convolution blocks with 32, 64, 128, and 256 filters	Three convolution blocks with 32, 64, and 128 filters
Recurrent refinement	Not used	Four ConvLSTM blocks with 192, 192, 128, and 128 filters
Shared features	Global average pooling plus 256 units	Dense layers of 512 and 256 units
Outputs	Four-class tumor output	Four-class tumor output and three-tier risk output
Optimization	Adam, fixed 0.001, ReduceLROnPlateau	Adam, cosine decay from 0.0005, augmentation, and class-dependent sample weights
Regularization	Dropout 0.30, 0.30, 0.40, 0.40, then 0.50 dense	Dropout 0.25, 0.25, 0.30, then 0.15 in each ConvLSTM block and 0.30 before shared dense layers
Parameters	457 156	7 832 007

0.95, 0.45, 0.30, and 0.00 are assigned to glioma, meningioma, pituitary, and no tumor, respectively. These weights reflect the relative malignant potential described by WHO CNS5 and the broad survival trends reported by CBTRUS [1], [2]. Because the score uses the full probability distribution, uncertain predictions between neighboring classes lead to an intermediate severity value instead of a sudden jump to another class. The score is

then grouped into four post-hoc tiers: Low for scores below 0.15, Moderate for scores from 0.15 to below 0.45, High for scores from 0.45 to below 0.70, and Critical for scores of 0.70 or higher.

Predictive entropy is calculated as $H = -\sum_c p_c \log p_c$, where p_c is the softmax probability for class c , and it is used as a single-pass measure of uncertainty [10]. Samples with entropy above 0.60 are flagged for review. The auxiliary risk head and the four-tier post-hoc severity score are reported separately.

Model performance is assessed on the established test dataset using accuracy, precision, recall, weighted F1, class-specific metrics, estimated 95% binomial confidence intervals, and one-vs-rest ROC analyses. Normalized confusion matrices are used to emphasize the main classification errors, and gradient-weighted class activation mapping (Grad-CAM) is applied to the final convolutional block to support qualitative evaluation [16].

V. EXPERIMENTAL RESULTS AND DISCUSSION

A. Overall and Class-Wise Performance

From the 1,600 reserved test images, 64 samples (4.0%) exceeded the entropy threshold and were flagged for expert review. The High tier produced the highest flag rate at 9.6%, while the Moderate tier contributed the most flagged samples because it covered more cases. Table III summarizes the main findings. Compared with the baseline, BrainRiskNet delivers 2.74 points higher accuracy and 2.78 points higher weighted

TABLE III
UNIFIED PERFORMANCE SUMMARY AND STAGED ABLATION PATH OF THE CNN BASELINE AND BRAINRISKNET. METRICS USE THE NOTATION SHOWN IN EACH PANEL.

Overall performance					
Model	Acc.	Prec.	Rec.	F1	CI
CNN baseline	88.38	88.73	88.38	88.20	± 1.57
BrainRiskNet class.	91.12	91.50	91.12	90.98	± 1.39
BrainRiskNet risk	92.50	—	—	92.33	± 1.29

Post-hoc severity tiers				
Tier	Samples	Acc.	Mean conf.	Flagged
Low	474	84.0	99.1	0.6
Moderate	695	93.1	96.5	6.2
High	114	93.0	94.8	9.6
Critical	317	99.1	98.4	2.2

Class-wise and risk tier performance						
Class or tier	CNN baseline			BrainRiskNet		
	P	R	F1	P	R	F1
Glioma	0.92	0.73	0.82	0.95	0.79	0.86
Meningioma	0.78	0.86	0.82	0.90	0.90	0.90
No tumor	0.90	0.98	0.94	0.86	0.99	0.92
Pituitary	0.94	0.97	0.95	0.95	0.96	0.95
Low risk	—	—	—	0.86	0.99	0.92
Medium risk	—	—	—	0.95	0.96	0.96
High risk	—	—	—	0.96	0.78	0.86

Ablation path			
Variant	Training setup	Acc.	Weighted F1
CNN baseline	No augmentation	88.38	88.20
CNN-ConvLSTM v1	Online augmentation; no sample weights	90.94	90.72
CNN-ConvLSTM v2 (BrainRiskNet)	Online augmentation + sample weights	91.12	90.98

Note: The ablations were staged rather than fully controlled, so the gains reflect the combined effects of the ConvLSTM backbone, online augmentation, and sample weighting.

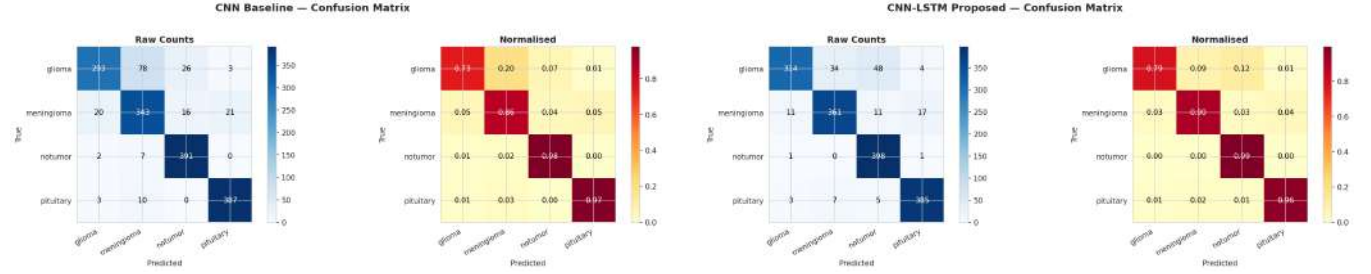


Fig. 2. Confusion matrices comparing the CNN baseline model (left) and the BrainRiskNet classification model (right). Each matrix shows both raw counts and normalized values, making the main error patterns easier to see.

F1, with the largest class-specific gains in glioma recall and meningioma precision and recall.

Fig. 2 shows that the remaining errors cluster primarily at the glioma–meningioma boundary, whereas no-tumor and pituitary cases are classified more consistently.

Grad-CAM examples in Fig. 3 show that correct predictions generally focus on lesion-containing regions rather than borders or background padding, supporting the use of meaningful image features.

Fig. 4 shows strong one-versus-rest discrimination across all four classes. Table III also shows that only 64 of 1600 test images (4.0%) exceeded the entropy threshold. The Critical tier was the most accurate and confident. By contrast, the lower accuracy in the Low tier (84.0%) is mainly associated with no-tumor cases that still received nontrivial probability mass for tumor classes, which raised the composite severity score above zero and placed them in the Low tier.

B. Comparison with Prior Work

Table IV places BrainRiskNet in the context of representative recent studies.

Compared with the three-class InceptionV3 benchmark of Rasheed et al. [7], BrainRiskNet reports higher summary

metrics, although the settings are not directly equivalent because the class definitions differ. The higher scores reported by Wong et al. [5] and Rasool et al. [8] are also not directly comparable because they rely on pretrained backbones and different dataset constructions or split protocols. Chauhan et al. [6] are included mainly for architectural context because their paper does not report the same summary metric set on the fixed split used here.

VI. LIMITATIONS

- The auxiliary risk labels are derived from tumor classes rather than patient outcomes, so the risk head should be viewed as a severity estimate rather than a prognosis model.
- Patient-level clinical data, such as age, symptom duration, molecular markers, and outcomes, were not available; therefore, all reported results are based only on image-derived predictions.
- All experiments use the Nickparvar Brain Tumor MRI Dataset, which combines images from Figshare, SARTAJ, and Br35H. Since the main public four-class brain tumor MRI datasets draw from the same source pool,

TABLE IV
COMPARISON WITH REPRESENTATIVE PRIOR STUDIES. VALUES MARKED WITH $\dagger$ ARE WEIGHTED AVERAGES DERIVED FROM PUBLISHED CLASS-WISE RESULTS, AND $\ddagger$ DENOTES AN F1-SCORE COMPUTED FROM REPORTED PRECISION AND RECALL.

Study	Setting	Acc.	Prec.	Rec.	F1	Pretrained	Classes	Comment
Wong et al. [5]	Four-class MRI, VGG16-style transfer-learning benchmark	99.24	99.08 $\dagger$	99.07 $\dagger$	99.30 $\dagger$	Yes	4	Aggregated dataset with ImageNet-pretrained VGG16; also reports 99.18 sensitivity and 99.75 specificity
Chauhan et al. [6]	BT-CNN hierarchical architecture	Not directly reported	—	—	—	No	4	Same dataset family, balanced binary tree architecture, evaluation protocol differs from present fixed split
Rasool et al. [8]	Hybrid GoogLeNet-SVM classifier	98.10	98.20	98.10	98.15 $\ddagger$	Yes	4	Different four-class dataset with hybrid transfer learning and random 80/20 split
Rasheed et al. [7]	InceptionV3, three-class benchmark	85.97	86.00	84.00	85.00	Yes	3	Three-class benchmark included as a coarse reference only
BrainRiskNet (this work)	Four classes, from-scratch public split	91.12	91.50	91.12	90.98	No	4	Adds entropy-based uncertainty flagging and an auxiliary risk head

testing across those datasets would not provide truly independent external validation. A prospective multi-institutional dataset with independently acquired MRI scans and patient-level outcome labels is still needed.

- A controlled comparison training both architectures under identical augmentation conditions was not performed; the reported accuracy gap therefore reflects the combined effect of the ConvLSTM architecture, augmentation, and class weighting rather than architecture alone.
- The system operates on two-dimensional slices rather than full three-dimensional volumes.
- The pseudo-sequence ConvLSTM module improves reshaped spatial representations instead of genuine volumetric temporal dynamics.

Together, these limitations mean that BrainRiskNet should be treated as an exploratory research system rather than a clinical tool.

VII. CONCLUSION

BrainRiskNet unifies four-class brain tumor MRI classification, auxiliary severity-oriented risk prediction, and entropy-based uncertainty flagging in a single from-scratch pipeline. On the fixed held-out set of 1600 images, the classification head reached 91.12% accuracy and 90.98% weighted F1, while the auxiliary risk head achieved 92.50% accuracy and 92.33% weighted F1. Together, these results show that a shared representation can support both diagnostic classification and a coarse severity-oriented research target.

The clearest gains appear in the clinically important glioma and meningioma classes, and the uncertainty signal offers a practical way to identify ambiguous cases for review. Overall, the study supports uncertainty-aware multitask modeling for brain tumor MRI analysis, while highlighting the need for external validation and outcome-grounded targets.

VIII. FUTURE WORK

Future work includes the following directions:

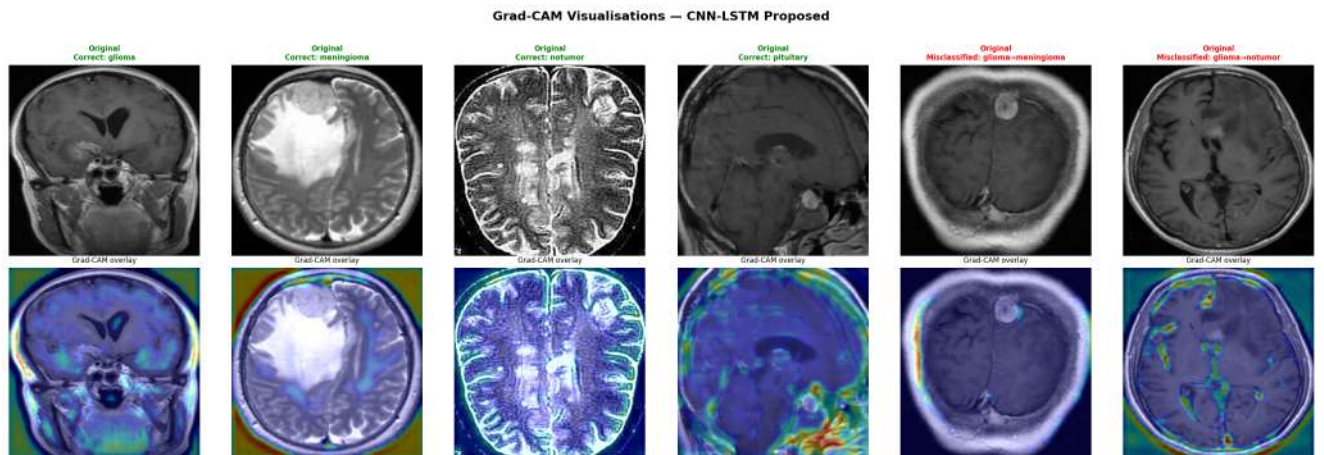


Fig. 3. Representative Grad-CAM visualizations for correctly classified glioma, meningioma, no-tumor, and pituitary cases, together with two representative misclassifications.

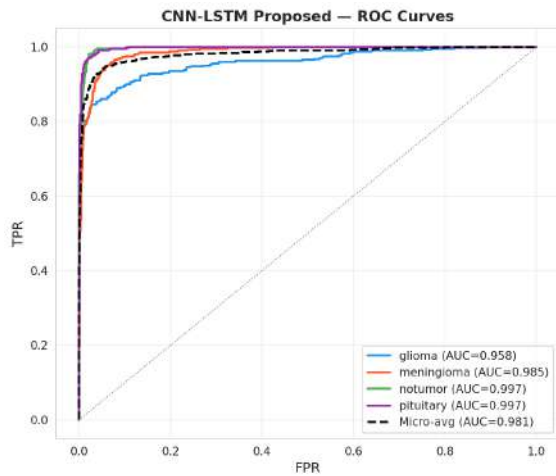


Fig. 4. ROC curves for the four tumor classes.

- 1) Models should move beyond isolated two-dimensional slices toward multi-slice or volumetric inputs, for example with three-dimensional convolutional encoders, volumetric ConvLSTM variants, or attention-based architectures.
- 2) Uncertainty estimation should be compared with Monte Carlo dropout, deep ensembles, and post-hoc calibration.

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